5 Numerical Methods



Joseph-Louis Lagrange
(25.01.1736 - 10.04.1813)

Introduction

Numerical Analysis is a branch of Mathematics which leads to approximate solution by repeated applications of four basic operations of Algebra. The knowledge of finite differences is essential for the study of Numerical Analysis.

Joseph-Louis Lagrange was an Italian mathematician and astronomer. he made significant contributions to the fields of analysis, number theory, and both classical and celestial mechanics.

Learning Objectives

After studying this chapter, students will be able to understand

- the finite differences
- how to find the polynomial using finite differences
- how to find the relations between the operators
- how to find the missing terms
- how to interpolate the values of a given series using Newton's interpolation formulae
- how to apply the Lagrange's interpolation formula

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5.1 Finite Differences

Consider the arguments $x_0, x_1, x_2, \cdots, x_n$ and the entries $y_0, y_1, y_2, \cdots, y_n$. $y = f(x)$ be a function of x. Let us assume that the values of x are in increasing order and equally spaced with a space length h. Then the values of x may be taken to be $x_0, x_0 + h, x_0 + 2h, \cdots, x_0 + nh$ and the function assumes the values $f(x_0), f(x_0 + h), f(x_0 + 2h), \cdots, f(x_0 + nh)$. Here we study some of the finite differences of the function $y = f(x)$.

5.1.1 Forward Difference Operator, Backward Difference Operator and Shifting Operator

Forward Difference Operator (Δ):

Let $y = f(x)$ be a given function of x. Let $y_0, y_1, y_2, \cdots, y_n$ be the values of y at $x = x_0, x_1, x_2, \ldots, x_n$ respectively. Then $y_1 - y_0,\ y_2 - y_1,\ y_3 - y_2, \ldots, y_n - y_{n-1}$ are called the first (forward) differences of the function y. They are denoted by $\Delta y_0, \Delta y_1, \Delta y_2, \ldots, \Delta y_{n-1}$ respectively.

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